

Capture the Flag 7: Stop Putting Everything on One Screen

Goal: Split your Student Directory into multiple screens using Flutter navigation, while using a modal for editing student information.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

Screenshot:

- Terminal showing your current branch

```
satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)
$ git status
On branch feature/conditional-ui
Your branch is up to date with 'origin/feature/conditional-ui'.

nothing to commit, working tree clean

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/conditional-ui)
$ git checkout -b feature/multi-screen-navigation
Switched to a new branch 'feature/multi-screen-navigation'
```

🚩 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Create a dedicated **Student List Screen** containing your existing student list.

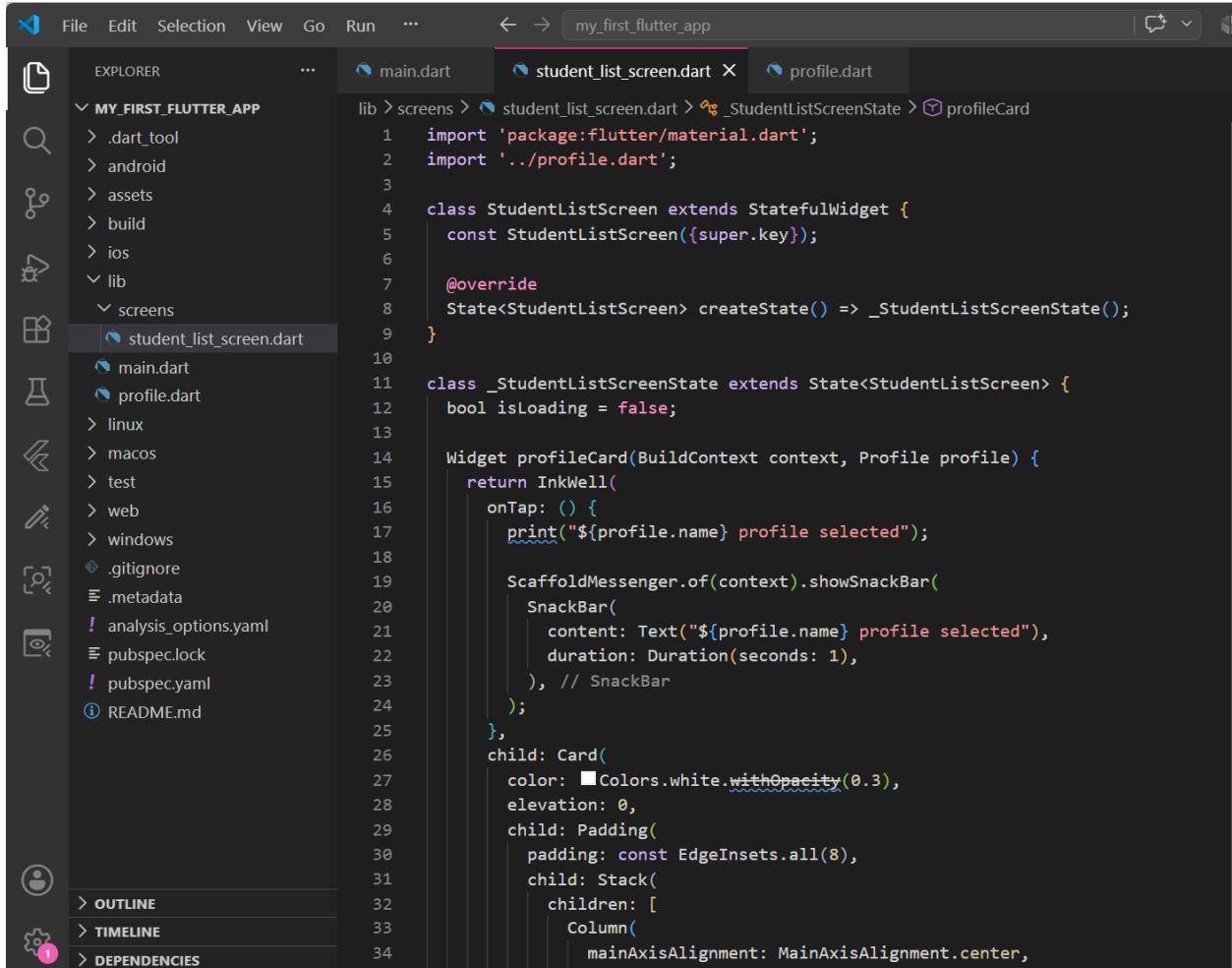
Research how Flutter creates separate screens using **Widget** classes.

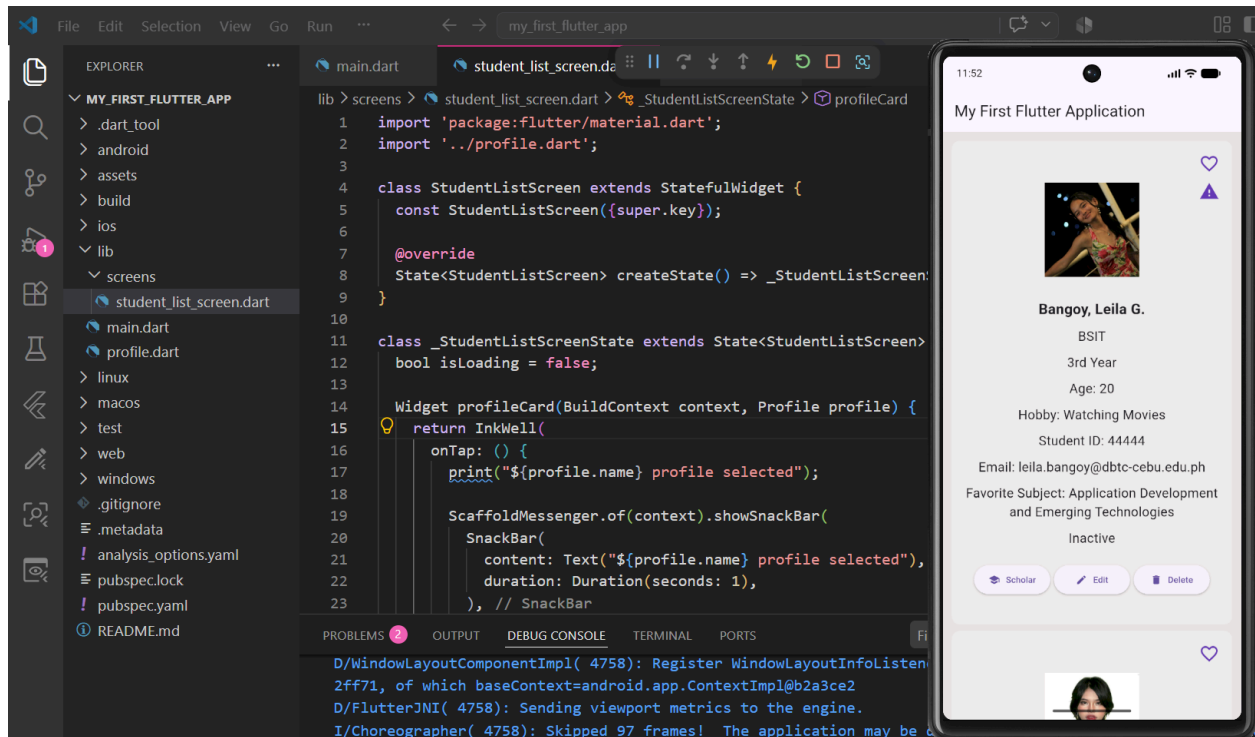
Test

The Student Directory should function as its own screen.

Screenshot:

- Student List Screen
- Student List screen code
- Application showing the screen





🚩 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

When a student is selected, navigate:

Student List → Student Details

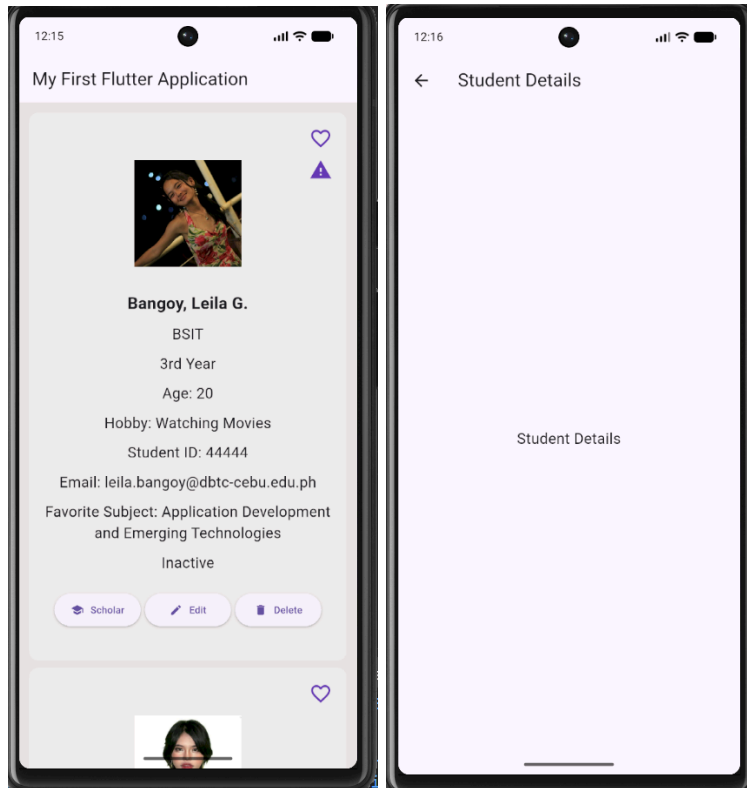
Research Flutter's **Navigator** and how to move between screens.

Test

1. Select a student.
2. Open the Student Details Screen. (No data displayed yet)
3. Return to the Student List using back navigation.

Screenshot:

- Student List Screen
- Student Details Screen
- Navigation code



```
17     onTap: () {
18         print("${profile.name} profile selected");
19
20         Navigator.push(
21             context,
22             MaterialPageRoute(
23                 builder: (context) {
24                     return const StudentDetailsScreen();
25                 },
26             ), // MaterialPageRoute
27         );
28     },
```

🚩 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

The Details Screen should display the correct information for the selected student.

Display relevant information such as:

- Name
- Student ID
- Course
- Status
- Other existing student data

Research how data can be passed between Flutter screens.

Test

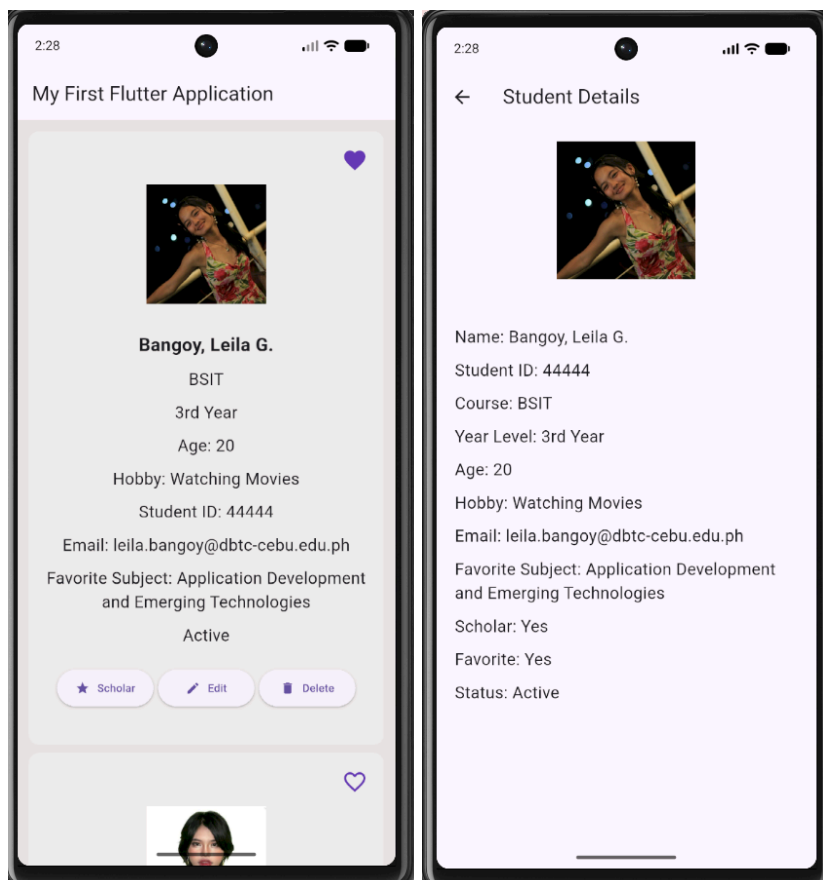
Selecting different students should display their corresponding information.

Student A → Student A's Details

Student B → Student B's Details

Screenshot:

- Selected student
- Student Details
- Data-passing code



```

15   Widget profileCard(BuildContext context, Profile profile) {
16       return InkWell(
17           onTap: () {
18               print("${profile.name} profile selected");
19           },
20           Navigator.push(
21               context,
22               MaterialPageRoute(
23                   builder: (context) {
24                       return StudentDetailsScreen(profile: profile);
25                   },
26               ), // MaterialPageRoute
27           );
28   },

```

```

20   class StudentDetailsScreen extends StatelessWidget {
21       final Profile profile;
22
23       const StudentDetailsScreen({super.key, required this.profile});

```

🚩 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Add an **Edit** action to the Student Details Screen.

When selected, display an editing interface using a **modal**, such as a dialog or bottom sheet.

Research Flutter's:

- `showDialog`
- `showModalBottomSheet`

You may choose whichever approach is appropriate for your design.

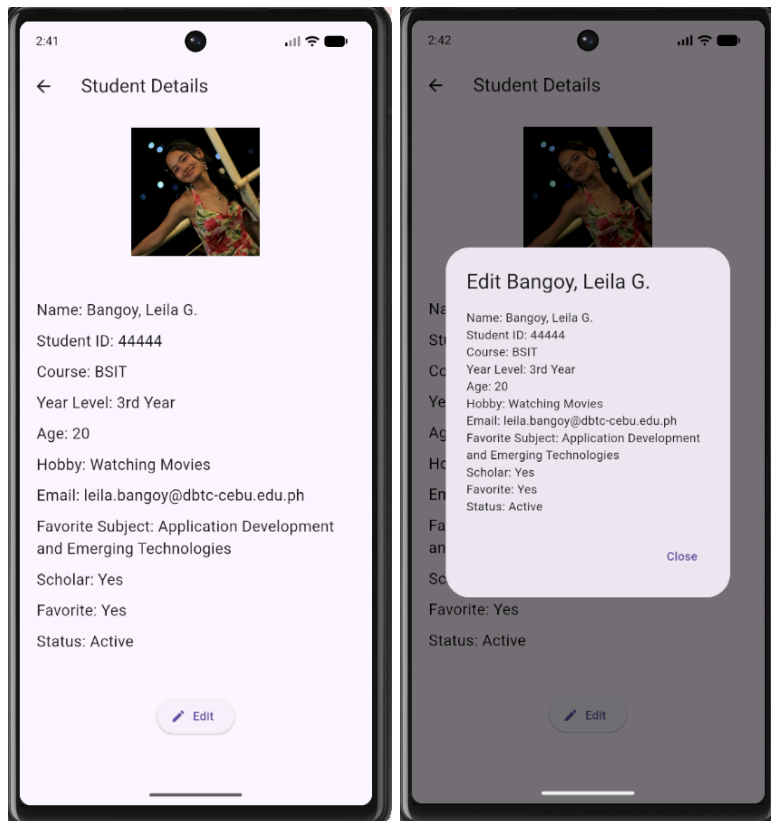
Test

1. Open Student Details.
2. Select Edit.
3. An editing modal should appear.
4. The modal should display the appropriate student's information.

5. Close or submit the modal.

Screenshot:

- Student Details Screen
- Edit modal
- Modal code




```

103 Center(
104   child: ElevatedButton.icon(
105     onPressed: () {
106       showDialog(
107         context: context,
108         builder: (context) {
109           return AlertDialog(
110             title: Text("Edit ${profile.name}"),
111             content: Text(
112               "Name: ${profile.name}\n"
113               "Student ID: ${profile.studentId}\n"
114               "Course: ${profile.course}\n"
115               "Year Level: ${profile.yearLevel}\n"
116               "Age: ${profile.age}\n"
117               "Hobby: ${profile.hobby}\n"
118               "Email: ${profile.email}\n"
119               "Favorite Subject: ${profile.favoriteSubject}\n"
120               "Scholar: ${profile.isScholar ? "Yes" : "No"}\n"
121               "Favorite: ${profile.isFavorite ? "Yes" : "No"}\n"
122               "Status: ${profile.active ? "Active" : "Inactive"}",
123             ), // Text
124             actions: [
125               TextButton(
126                 onPressed: () => Navigator.pop(context),
127                 child: const Text("Close"),
128               ), // TextButton
129             ],
130           ); // AlertDialog
131         },
132       );
133     },
134     icon: const Icon(Icons.edit),
135     label: const Text("Edit"),
136   ), // ElevatedButton.icon
137 ), // Center

```

Flag 5 — Create the Add Student Screen

Objective: Create a separate screen for adding a new student.

Add an action from the Student List that navigates to:

Student List → Add Student

Create the **Add Student Screen**.

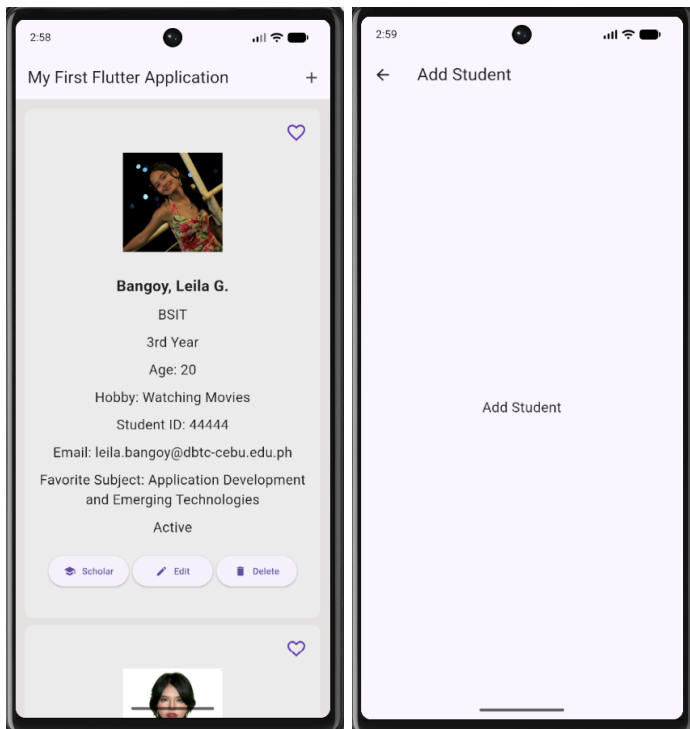
For now, focus on creating the screen and navigation structure.

Test

Selecting Add Student should open the Add Student Screen.

Screenshot:

- Student List with Add action
- Add Student Screen
- Navigation code



```
241     appBar: AppBar(  
242       title: const Text("My First Flutter Application"),  
243       actions: [  
244         IconButton(  
245           onPressed: () {  
246             Navigator.push(  
247               context,  
248               MaterialPageRoute(  
249                 builder: (context) {  
250                   return const AddStudentScreen();  
251                 },  
252               ), // MaterialPageRoute  
253             );  
254           },  
255           icon: const Icon(Icons.add),  
256         ), // IconButton  
257       ],  
258     ), // AppBar
```

🚩 Flag 6 — Prepare the Add Student Screen

Objective: Prepare the Add Student Screen for future functionality.

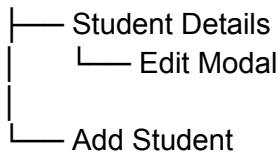
Organize the screen so it is ready to receive student input.

Do not implement the student input fields yet.

The screen should simply exist as a separate destination.

Your application should now have:

Student List

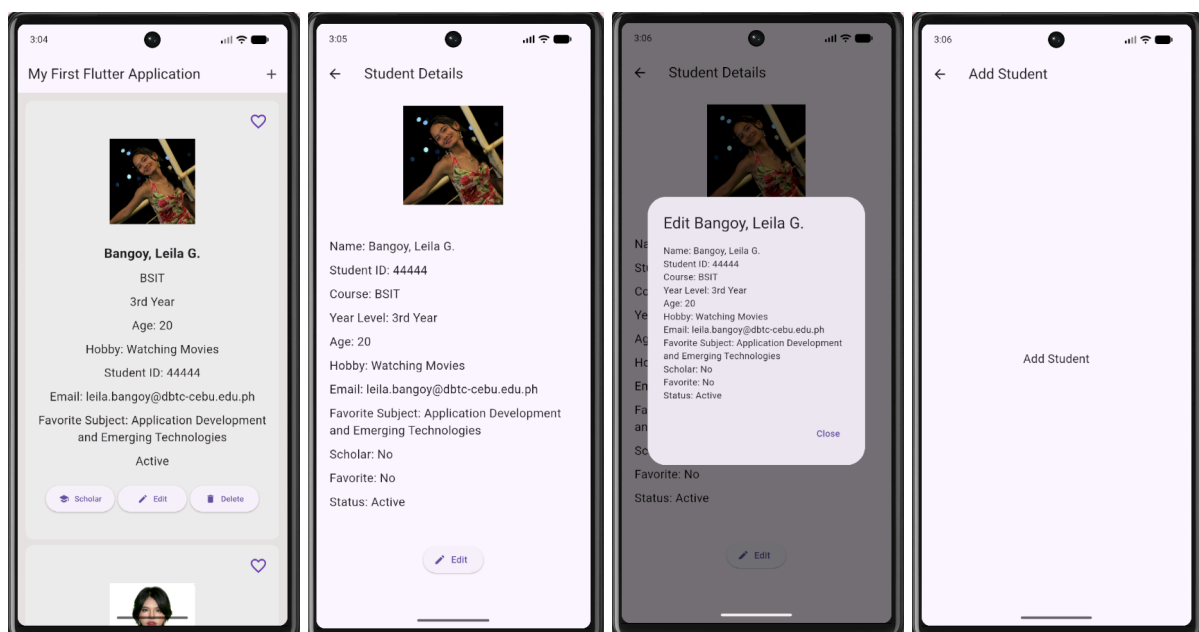


Test

You should be able to navigate between the appropriate screens without placing everything on one screen.

Screenshot:

- Student List
- Student Details
- Edit Modal
- Add Student Screen



🚩 Flag 7 — Connect the Routes

Objective: Organize the application's screens using Flutter routes.

Create routes for the major screens:

/student-list
/student-details
/add-student

Research:

- Named routes
- `Navigator.pushNamed`
- Passing arguments through routes

The **Edit Modal** should not be a route.

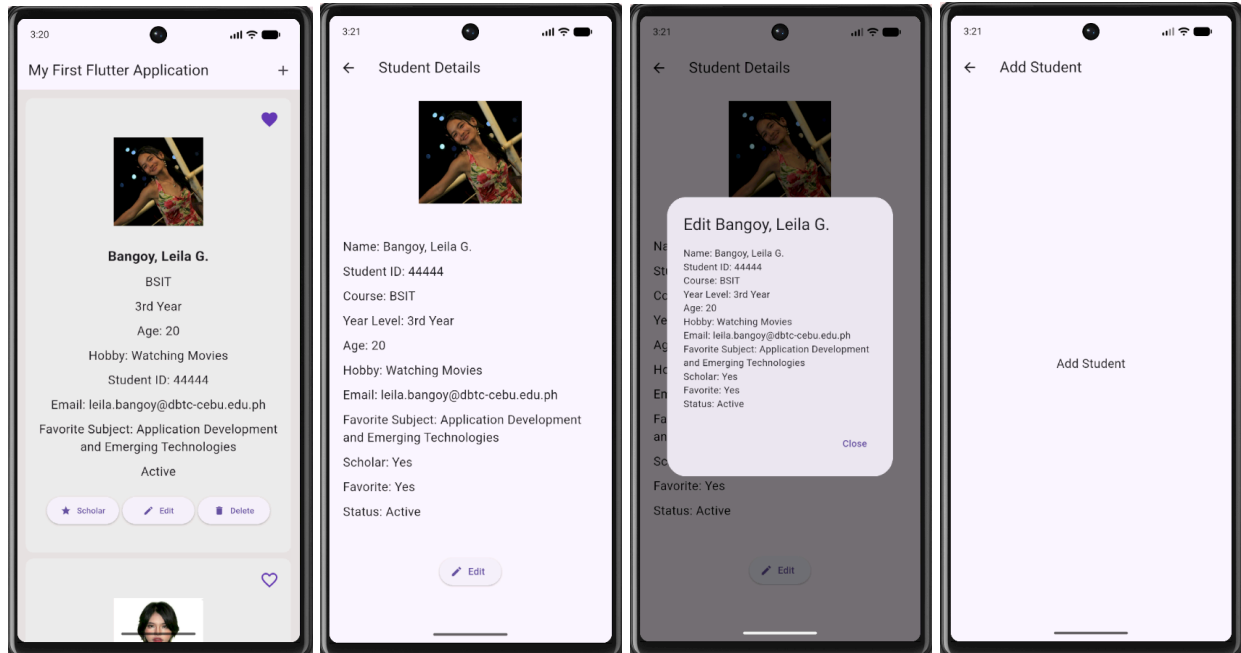
Test

Verify that:

- Student List opens correctly.
- Student Details opens correctly.
- The correct student data is passed.
- Add Student opens correctly.
- Edit opens as a modal from Student Details.

Screenshot:

- Route configuration
- Navigation using routes
- Student Details receiving student data
- Edit modal



```

21     initialRoute: '/student-list',
22
23     routes: {
24       '/student-list': (context) {
25         return const StudentListScreen();
26       },
27
28       '/student-details': (context) {
29         final profile = ModalRoute.of(context)!.settings.arguments as Profile;
30         return StudentDetailsScreen(profile: profile);
31       },
32
33       '/add-student': (context) {
34         return const AddStudentScreen();
35       },
36     },

```

```

19     Navigator.pushNamed(context, '/student-details', arguments: profile);
20   },

```

```

236   onPressed: () {
237     Navigator.pushNamed(context, '/add-student');
238   },

```

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

feature/multi-screen-navigation

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
MINGW64~/c/Users/satom/OneDrive - TAFE NSW/IT314AB_Verano

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-scre
en-navigation)
$ git branch
  feature/conditional-ui
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/multi-screen-navigation
  feature/student-interactions
  feature/student-list
  feature/widgets
  main

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-scre
en-navigation)
$ git status
On branch feature/multi-screen-navigation
Your branch is up to date with 'origin/feature/multi-screen-navigation'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   FlutterProjects/my_first_flutter_app/lib/main.dart
        modified:   FlutterProjects/my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc
        modified:   FlutterProjects/my_first_flutter_app/linux/flutter/generated_plugin_registrant.h
        modified:   FlutterProjects/my_first_flutter_app/linux/flutter/generated_plugins.cmake
        modified:   FlutterProjects/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift
        modified:   FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc
        modified:   FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h
        modified:   FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugins.cmake

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        FlutterProjects/my_first_flutter_app/lib/profile.dart
        FlutterProjects/my_first_flutter_app/lib/screens/

no changes added to commit (use "git add" and/or "git commit -a")
```

```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-screen-navigation)
$ git add
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterProjects/my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-screen-navigation)
$ git commit -m "feat: submit Stop Putting Everything on One Screen"
error: pathspec '-' did not match any file(s) known to git
error: pathspec 'm' did not match any file(s) known to git
error: pathspec 'feat: submit Stop Putting Everything on One Screen' did not match any file(s) known to git

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-screen-navigation)
$ git commit -m "feat: submit Stop Putting Everything on One Screen"
[feature/multi-screen-navigation 1fe9caf] feat: submit Stop Putting Everything on One Screen
5 files changed, 545 insertions(+), 354 deletions(-)
create mode 100644 FlutterProjects/my_first_flutter_app/lib/profile.dart
create mode 100644 FlutterProjects/my_first_flutter_app/lib/screens/add_student_screen.dart
create mode 100644 FlutterProjects/my_first_flutter_app/lib/screens/student_details_screen.dart
create mode 100644 FlutterProjects/my_first_flutter_app/lib/screens/student_list_screen.dart

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-screen-navigation)
$ git status
On branch feature/multi-screen-navigation
Your branch is ahead of 'origin/feature/multi-screen-navigation' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/multi-screen-navigation)
$ git push
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 20 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 2.55 KiB | 2.55 MiB/s, done.
Total 11 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 3 local objects.
To https://github.com/KVerano/IT314AB_Verano.git
  74ecf3e..1fe9caf  feature/multi-screen-navigation -> feature/multi-screen-navigation

```